

CHAPTER 9

Use and Importance of Personality Variables in Work Settings

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USE AND IMPORTANCE OF PERSONALITY VARIABLES IN WORK SETTINGS

Personality variables, once thought unimportant for understanding work behavior, are now widely accepted and used in research and applied work settings. Nuanced thinking and research have produced empirical evidence demonstrating the importance of personality variables as determinants of work behavior and performance for individuals, teams, and organizations. Human behavior, whether in a work or nonwork setting, is complicated, with multiple determinants. It is not easily predicted or explained. Nonlinear relationships, long hypothesized, are now being documented and gaining acceptance. Relationships between personality variables and work-related outcomes are complex and new process models of determinants of work behavior better reflect that complexity.

Issues that once bedeviled research on and use of personality variables in work settings are yielding to well-conceived research questions and designs. Intractable measurement issues are succumbing to intense research efforts that have produced insights and advances. The decade between the first edition of the *Handbook of Psychology* and this, the second edition, has seen a substantial amount of research and significant progress. Nonetheless, some issues still haunt.

This chapter describes what we have learned in the past decade and integrates it with what we learned before that has stood the test of time. Separate sections focus on (a) the structure of personality variables (traits) and why we care about structure; (b) models of the determinants of work performance that include personality variables; (c) construct validity of personality variables, including the magnitude of criterion-related validity (effect size) of personality variables for predicting work-related outcomes; (d) effects of including personality measures for hiring and promotion decisions on demographic and ethnic groups; (e) advantages and disadvantages of different methods of measuring personality variables; and (f) effects of intentional distortion (especially self-report measures) on validity and strategies for ameliorating its effects. We close with a summary of the evidence and suggestions for a path forward.

STRUCTURE OF PERSONALITY

In early personality research, correlations were computed between all personality variables and all criteria. Most of these correlations were near zero, creating the impression that personality was generally unrelated to performance. We now better understand that the trait being investigated must be relevant to the criterion, and that predictors and

criteria should be conceptualized as constructs (Hough & Schneider, 1996). Meta-analyses of the criterion-related validity of personality variables have illustrated the benefit of using personality taxonomies as an organizing framework, revealing personality–performance relationships that had not been clear before (e.g., Barrick, Mount, & Judge, 2001; Hough, Eaton, Dunnette, Kamp, & McCloy, 1990; Hough & Furnham, 2003). Research has shown that lower level facets of the same Big Five factor often have very different correlations with job performance criteria, revealing meaningful relationships that are masked if broader measures are used (Hough, 1992). Thus, linking specific predictor and criterion measures can result in increased correlations and enhanced understanding of the relationship between personality and performance. The linking of these lower level predictors and criteria requires taxonomies of specific personality and job performance constructs (Barrick et al., 2001; Hough, 1992; Hough, 2003).

The dominant approach to identifying a structure of personality has been the lexical approach, which is based on the words people use to describe each other. The Big Five (also known as the Five-Factor Model, or FFM) is an example of a personality structure derived using the lexical approach. The FFM has been ubiquitous in personality research, being robust and generalizable across rating sources, some cultures, some languages, and factor extraction and rotation methods (Hough & Furnham, 2003). The five factors are generally labeled Extraversion, Conscientiousness, Agreeableness, Emotional Stability, and Openness to Experience (Digman, 1990).

More recent research has found less consistency across studies in the robustness of the FFM (see Oswald & Hough, 2011, for a review of this research). Ashton et al. (2004) reanalyzed the data from studies examining the factor structure of lexical terms across seven languages (Dutch, French, German, Hungarian, Italian, Korean, and Polish) and found a sixth factor in addition to analogs of each of the FFM factors. They labeled this sixth factor Honesty/Humility and the six-factor model was named the HEXACO model. The six-factor solution did a better job than the FFM of accounting for the variance in both lexical ratings across languages and predictor–criterion relationships. For example, across four cross-cultural samples, the six factors account for 10 to 15% more variance in the prediction of workplace delinquency than did the FFM (Lee, Ashton, & deVries, 2005).

Although the FFM has advanced theory and practice as a useful framework for organizing and summarizing personality–performance relationships, it has been

criticized for being insufficiently comprehensive and too heterogeneous (Block, 1995; Hough & Dilchert, 2010; Hough & Oswald, 2008; Hough & Schneider, 1996; Oswald & Hough, 2011; Paunonen & Jackson, 2000; Schneider, Hough, & Dunnette, 1996). Research shows that the more specific facets that make up the broad factors of the FFM and HEXACO models differ in their relationships with important work-related criteria (Ashton, 1998; Dudley, Orvis, Lebiecki, & Cortina, 2006; Moon, Hollenbeck, Marinova, & Humphrey, 2008; Paunonen & Nicol, 2001; Roberts, Chernyshenko, Stark, & Goldberg, 2005; Vinchur, Schippmann, Switzer, & Roth, 1998; Warr, Bartram, & Martin, 2005). In addition, subgroup (e.g., race, gender, age) mean differences have been found for some facets within a factor but not others, or subgroup differences are in the opposite direction for facets within the same factor (Foldes, Duehr, & Ones, 2008; Hough, Oswald, & Ployhart, 2001). Therefore, measuring personality at the narrower facet level can often provide more useful information than measurement at the broader factor level. Moreover, scores at the facet level can be aggregated to form more heterogeneous constructs at the five- and six-factor levels. Even higher level, more heterogeneous, or compound variables such as Managerial Potential or Social Service Orientation can be formed from facet-level scales.

The number of lower order facets of the FFM is still open to debate, but Saucier and Ostendorf (1999) provided a good starting point by identifying 18 subcomponents of the FFM that were replicable across two languages (English and German). These subcomponents were labeled sociability, unrestraint, assertiveness, activity–adventurousness (all facets of Extraversion), warmth–affection, gentleness, generosity, modesty–humility (Agreeableness), orderliness, decisiveness–consistency, reliability, industriousness (Conscientiousness), irritability, insecurity, emotionality (Emotional Stability), intellect, imagination–creativity, and perceptiveness (Openness to Experience).

An alternative to developing a personality taxonomy based on factor analysis of intercorrelations between scores on personality variables is a nomological–web clustering approach, in which taxons are based on similarities in patterns of relationships with variables outside the personality domain, such as job performance criteria (Hough & Ones, 2001; Hough & Furnham, 2003). The nomological–web clustering approach is based on the idea that personality variables that are grouped together should have similar patterns of correlations with other variables, which is consistent with demonstrating convergent and

discriminant validity to support construct validity (Campbell & Fiske, 1959). Based on an extensive review of the literature, Hough and Ones (2001) used this approach to propose a working taxonomy of personality variables and called for other researchers to continually refine this taxonomy through theory and empirical evidence. This taxonomy has since been employed by Dudley et al. (2006) and Foldes et al. (2008) to contribute to a greater understanding of the relationships between personality variables and work-related criteria.

A related approach is the use of compound traits, which are combinations of basic personality traits that do not necessarily covary, but are combined into a single variable to maximize the prediction of a specific criterion construct (Hough & Schneider, 1996). Some examples of compound personality traits are integrity (Ones, Viswesvaran, & Schmidt, 1993), customer service orientation (Frei & McDaniel, 1998), employee reliability (Hogan & Hogan, 1989), and managerial potential (Gough, 1984). Hough and Ones (2001) suggested a number of other possible compound traits.

MODELS OF DETERMINANTS OF WORK PERFORMANCE

The past decade has seen a surge in the development of models of the process by which personality influences job performance, in the form of searching for moderators of the relationship between personality and performance and searching for mediators of this relationship. This type of research focuses on understanding in greater depth the nature of personality, job performance, and how they are linked. To account for a substantial proportion of the variance in the criterion space, we need theories and models that reflect the complexity of how personality influences the determinants of job performance. A number of process models of how personality influences performance have been proposed in recent years, both for individual job performance and for counterproductive work behavior (CWB).

Johnson (2003) built on and expanded earlier performance prediction models proposed by Campbell (1990) and Motowidlo, Borman, and Schmit (1997), focusing on the role of personality in determining an individual's level of performance on specific performance dimensions. In these models, performance is determined directly by a combination of knowledge, skill, and motivation, to which Johnson added work habits as a fourth determinant that is distinct from motivation. These direct determinants are

distinguished from indirect determinants, which can only influence performance via the direct determinants. Personality is an indirect determinant, along with variables such as abilities, education, and experience. Johnson incorporated Mitchell and Daniels' (2003) conceptualization of motivation, which distinguishes between proactive cognitive processes (e.g., expectancies, self-efficacy, goal setting) and online cognitive processes (e.g., self-regulation). Johnson added psychological motives (e.g., values, interests, preferences, attitudes) as a third component of motivation that may mediate the relationship between personality and proactive cognitive processes. Johnson, Duehr, Hezlett, Muros, and Ferstl (2008) found partial support for this model for five different performance dimensions. As Heckhausen and Kuhl (1985) suggested, personality variables had differential relationships with different components of motivation. This finding, along with direct paths to performance from each motivation component, supports the need for splitting motivation into its components to truly understand how personality influences performance through motivation.

Barrick, Mitchell, and Stewart's (2003) model is consistent with Johnson's (2003) model, but they focused exclusively on the influence of motivation while recognizing the importance of situational demands on the way personality is expressed. In this model, three types of motivation (accomplishment striving, status striving, and communion striving) mediate the relationship between personality and performance. The relationship between a given personality trait and behavior is stronger when the situation is relevant to the trait's expression and allows enough autonomy that individual behavior varies. Barrick et al. focused on cooperative and competitive social demands as situational aspects that influence trait expression. The relationship between accomplishment striving and performance is mediated by communion striving in situations in which cooperative demands operate and by status striving in situations in which competitive demands operate. Barrick, Stewart, and Piotrowski (2002) found partial support for this model, showing that status striving mediated the relationship between accomplishment striving and performance in a sample of sales representatives who worked in a competitive sales setting.

Tett and Burnett's (2003) trait activation theory describes the influence of the situation on trait expression in more detail. According to this theory, traits are activated by cues, which are situations that provide trait expression opportunities. Trait-consistent behavior is considered job performance when the behavior is valued by the organization. Behavior is reinforced by the intrinsic

satisfaction associated with expressing your traits and the extrinsic satisfaction from being rewarded for good performance. Trait activation theory predicts that the correlation between a personality trait and job performance depends on the extent to which the situation provides cues for trait expression that are valued by those who evaluate performance. Positive correlations are expected when trait expression is considered to meet work demands, and negative correlations are expected when trait expression is considered to interfere with meeting work demands. Predictions made by trait activation theory have been supported in some field studies (De Hoogh, Den Hartog, & Koopman, 2005; Kamdar & Van Dyne, 2007; Li, Liang, & Crant, 2010; Lievens, Chasteen, Day, & Christiansen, 2006).

Some studies have found that job satisfaction mediates the relationship between personality and job performance. Ilies, Fulmer, Spitzmuller, and Johnson (2009) found that job satisfaction partially mediated the relationship between personality (Agreeableness and Conscientiousness) and organizational citizenship behavior (OCB). Agreeableness had both direct and indirect effects through job satisfaction on OCB directed to individuals, while Conscientiousness had both direct and indirect effects through job satisfaction on OCB directed to the organization. Mount, Ilies, and Johnson (2006) found similar results when testing job satisfaction as a mediator between those same personality traits and counterproductive work behavior (CWB). Johnson et al. (2008) found that job satisfaction mediated the relationship between personality variables and the performance dimensions of maintaining good working relationships and organizational commitment, which are related to CWB and OCB directed to the individual and to the organization.

Van Iddekinge, Ferris, and Heffner (2009) found support for a multistage model of leader performance in which knowledge, skills, and ability were direct determinants of performance, and the indirect determinants were three personality traits—cognitive ability, leadership experience, and leadership motivation. In this model, leadership experience and motivation were more proximal antecedents than were personality and cognitive ability, and they partially mediated the relationship between the more distal determinants and the direct determinants.

Examination of these process models reveals that they are generally consistent with each other, with each model emphasizing different aspects of the personality–performance process. Johnson and Hezlett (2008) integrated a number of process models into a single high-level model in which all variables are expressed as broad

construct domains and all of the major potential influences of personality on performance are incorporated. The purposes of the general model are to (a) provide a guide for constructing separate models for specific performance components, and (b) identify potential paths through which personality influences performance. Because this is a general model of potential influences on performance, the model for any specific performance dimension would not necessarily include each potential path. Research is necessary on specific performance dimensions to determine what elements of the general model operate for different types of performance. This model highlights the many potential routes through which personality may influence job performance. Personality is included as a determinant of every direct determinant of performance, as well as other indirect determinants such as motivational variables and attitudes. In addition, personality is a potential moderator of many of the relationships specified in this model.

Johnson and Schneider (in press) review a number of personality process models and how they fit into Johnson and Hezlett's (2008) general model. They also suggest a strategy for moving research on personality process models forward in a more systematic way by pitting competing theories against each other and shifting the research focus from confirmation to disconfirmation.

CONSTRUCT VALIDITY OF PERSONALITY VARIABLES

A recurring criticism of personality measures as selection instruments is that the magnitude of criterion-related validity (effect size) of personality variables for predicting work-related outcomes is too small to matter. Although 10 years ago, in the first edition of the *Handbook of Psychology*, we thought we had provided evidence of the usefulness of personality variables for predicting important work-related outcomes, this criticism surfaced again. (See Morgeson et al., 2007, for the criticism; see Ones, Dilchert, Viswesvaran, & Judge, 2007, and Tett & Christiansen, 2007, for counterarguments.) At that time, we (Hough & Furnham, 2003) summarized meta-analytically derived criterion-related validities of personality constructs according to criterion constructs and work-related outcomes. Our observations at that time were:

First, validities vary within cells, some of which is due to methodological differences in the studies. For example, some of the variation within cells is due to the number and types of corrections made to observed validities. . . . (p. 154)

Second, and perhaps more important, the validities of personality variables vary according to criterion construct. Although few meta-analyses are conducted at the facet level, a comparison of the validities of facets reveals interesting differences at that level as well—some facets from the same Big Five factor show different patterns of correlations with criterion constructs. . . . More researchers should report validities at the facet level, thus enabling meta-analysis of relationships at that level. (p. 155)

Third, compared to other Big Five variables, Conscientiousness correlates most highly with overall job performance, a conclusion similar to that of Barrick et al. (2001). However, depending upon the job, conscientiousness facets—achievement and dependability—are differentially important. . . . (p. 155)

Fourth, although Conscientiousness correlates most highly with overall job performance for many jobs, the validities of other personality variables are often higher for specific performance constructs. . . . (p. 155)

Fifth, compound variables often show the highest levels of validity when the criterion is complex. . . . (p. 155)

We have compiled a large majority of the meta-analyses that now exist of criterion-related validities of personality variables (facet- and broader-level FFM factors as well as compound variables) for predicting work-related criteria. The results appear in Tables 9.1 through 9.6. With dozens more meta-analyses completed in the past decade, these tables contain many more variables on both the predictor and the criterion side than were contained in the Hough and Furnham (2003) tables. With more variables and larger sample sizes, the conclusions drawn before are still appropriate but we can have more confidence in them.

In addition, having data on such a variety of predictors and criteria allows us to examine patterns of validities for different types of personality variables and different criteria. In general, validities tend to be high when predictor and criterion are conceptually matched and low when they are not. For example, Sales Effectiveness is predicted strongly by Achievement and Dominance, and to a lesser extent by Dependability, Sociability, and Rugged Individualism. It is not predicted by Emotional Stability, Agreeableness, and Openness to Experience. This is what one would expect for this criterion. Similar patterns are found when examining the tables for other criteria.

The overall conclusion is that personality variables are highly useful for predicting a wide variety of criteria of interest to organizational researchers. We cannot expect all personality variables to predict all criteria, but prediction

can be very strong when the predictor construct and the criterion construct are conceptually coherent.

SYNTHETIC VALIDITY

Barrick et al. (2001) quantitatively summarized 15 meta-analytic studies relating the FFM to job performance conducted prior to 2001, which they suggested summarized “what we have learned about personality-performance relationships over the past century” (p. 9). They called for a moratorium on meta-analytic studies of the type they reviewed and suggested, among other things, that researchers should embark on a new research agenda focused on linking lower-level predictors and criteria. As we reviewed earlier, more specific personality facets have higher correlations with performance criteria than do broader traits such as those in the FFM. Around the same time, Hough (2001) called for the development of a database linking predictor and criterion constructs that can be used with synthetic validation models to build prediction equations for specific situations. Recently, Johnson and colleagues (2010) again called for the creation of this type of database, arguing that synthetic validation, in conjunction with a database linking specific predictors and criteria, has the potential to substantially advance the science and practice of industrial and organizational psychology.

We believe that the application of personality constructs to personnel selection can be advanced significantly through synthetic validation. Synthetic validation is a logical process of inferring validity on the basis of the relationships between components of a job (i.e., clusters of similar tasks or work behaviors) and tests of the attributes that are needed to perform those components (Mossholder & Arvey, 1984). It can be applied in situations in which multiple jobs share a number of the same job components, such that relationships can be identified between predictors and job components across jobs with larger sample sizes than can be obtained within jobs. According to Johnson et al.’s (2010) vision, when a synthetic validation database has been developed to a significant extent, practitioners will be able to buy or develop measures of predictor constructs that have been shown to predict performance on job components relevant to any job of interest and to calculate a validity coefficient for that job. The database will also advance science by greatly increasing our knowledge base with respect to relationships between different predictor and criterion constructs. Database development will lead to much quicker accumulation of this information than would otherwise occur.

TABLE 9.1 Meta-analytic Criterion-Related Validities of Self-Report Conscientiousness and Its Facets for Predicting Work Outcomes

Work Outcome (Criterion Construct)	Conscientiousness Facet				Global Conscientiousness
	Cautiousness/Impulse Control	Order	Dependability	Achievement	
Overall (Individual) Job Performance	$\rho_{pc} = -.01^l(k = 22; N = 2,753)$	$\rho_{pc} = .10^l(k = 26; N = 3,835)$	$r = .20^f(k = 9; N = 4,362)$ $\rho_{pc} = .25^l(k = 15; N = 1,748)$ $r = .07^p(k = 114; N = 21,029)$	$r = .18^f(k = 9; N = 4,362)$ $\rho_{pc} = .20^l(k = 26; N = 3,595)$ $r = .19^p(k = 31; N = 3,182)$	$\rho = .23^b(k = 92; N = 12,893)$ $\rho = .24^s(k = 42; N = 7,342)$ $\rho = .16^{pp}(k = 18; N = 2,241; \text{Europe})$ $\rho_{rc} = .28^{rr}(k = 90; N = 19,460; \text{FFM scales})$ $\rho_{pc} = .31^k(k = 5; N = 203)$ $\rho_{pc} = .42^{mm}(k = 5; N = 266; \text{professional teams})$ $\rho_{pc} = .00^{mm}(k = 5; N = 261; \text{student teams})$
Overall (Team) Performance					
Specific Occupation Success:					
Sales Effectiveness	$\rho_{pc} = -.04^l(k = 7; N = 705)$	$\rho_{pc} = .19^l(k = 5; N = 849)$	$\rho_{pc} = .26^l(k = 5; N = 592)$ $r = .06^p(k = 5; N = 2,236)$ $\rho_{rc} = .18^{uu}(k = 15; N = 1,702; \text{ratings})$ $\rho_{rc} = .18^{uu}(k = 5; N = 359; \text{objective sales})$	$\rho_{pc} = .28^l(k = 4; N = 532)$ $r = .27^p(k = 2; N = 162)$ $\rho_{rc} = .25^{uu}(k = 8; N = 1,319; \text{ratings})$ $\rho_{rc} = .41^{uu}(k = 10; N = 1,269; \text{objective sales})$	$\rho = .23^b(k = 21; N = 2,263)$ $\rho = .29^s(k = 10; N = 1,369)$ $\rho_{rc} = .21^{uu}(k = 19; N = 2,186; \text{ratings})$ $\rho_{rc} = .31^{uu}(k = 15; N = 1,774; \text{objective sales})$
Skilled/Semiskilled	$\rho_{pc} = -.20^l(k = 4; N = 273)$	$\rho_{pc} = .21(k = 8; N = 987)$	$\rho_{pc} = .27^l(k = 3; N = 376)$	$\rho_{pc} = .20^l(k = 7; N = 766)$	$\rho = .21^b(k = 25; N = 4,588)$ $\rho = .17^s(k = 14; N = 3,481)$
Customer Service		$\rho_{pc} = .12^l(k = 5; N = 659)$		$\rho_{pc} = .22^l(k = 5; N = 659)$	$\rho = .27^s(k = 12; N = 1,849)$ $\rho = .29^{ec}(k = 7; N = 908)$
Managerial Effectiveness	$\rho_{pc} = .01^l(k = 7; N = 1,348)$	$\rho_{pc} = -.12^l(k = 6; N = 1,148)$	$\rho_{pc} = .19^l(k = 3; N = 402)$ $\rho = .03^r(k = 62; N = 5,078)$	$\rho_{pc} = .13^l(k = 7; N = 1,330)$ $\rho = .17^r(k = 78; N = 11,926)$	$\rho = .22^b(k = 52; N = 10,058)$ $\rho = .11^r(k = 186; N = 50,367)$ $\rho = .19^s(k = 4; N = 495)$

Entrepreneurship—Business Creation	$\rho_{pc} = -.10^{oo} (k = 18; N = 8,863)$			$\rho_{pc} = .22^{oo} (k = 29; N = 8,698)$	
Entrepreneurial Success	$\rho_{pc} = -.10^{oo} (k = 13; N = 1,744)$			$\rho_{pc} = .30^{oo} (k = 31; N = 4,115)$	
Expatriate Effectiveness					$\rho_{pc} = .17^{bb} (k = 11; N = 1,023)$
Combat Effectiveness			$r = .08^p (k = 5; N = 1,490)$		
Work Success Components:					
Contextual/Organizational Citizenship (OCBs)—Overall					$r = .19^g (k = 10; N = 1,963)$
OCBs—Dedication	$\rho_{pc} = .08^l (k = 6; N = 878)$	$\rho_{pc} = .10^l (k = 13; N = 1,658)$	$\rho_{pc} = .46^l (k = 7; N = 979)$	$\rho_{pc} = .39^l (k = 15; N = 2,330)$	$\rho_{pc} = .23^z (k = 15; N = 848)$
OCBs—Interpersonal Facilitation	$\rho_{pc} = .00^l (k = 12; N = 2,187)$	$\rho_{pc} = -.02^l (k = 18; N = 3,011)$	$\rho_{pc} = .23^l (k = 4; N = 627)$	$\rho_{pc} = .11^l (k = 18; N = 3,264)$	$\rho = .18^s (k = 23; N = 4,301)$
OCBs—General Compliance					$\rho_{pc} = .23^{ll} (k = 7; N = 1,231)$
OCBs—Altruism					$\rho_{pc} = .04^{ll} (k = 7; N = 1,231)$
Counterproductive Work Behavior (CWBs)—Overall	$\rho_{pc} = -.11^l (k = 14; N = 2,251)$	$\rho_{pc} = -.07^l (k = 13; N = 1,311)$	$r = -.21^f (k = 9; N = 4,362)$ $\rho_{pc} = -.34^l (k = 16; N = 2,195)$ $r = -.24^p (k = 69; N = 98,676)$	$r = -.08^f (k = 9; N = 4,362)$ $\rho_{pc} = .00^l (k = 13; N = 1,026)$ $r = -.19^p (k = 4; N = 19,476)$	$\rho_{pc} = -.35^d (estimated)$
CWBs—Interpersonal Deviance					$\rho_{pc} = -.23^d (k = 11; N = 3,458)$
CWBs—Organizational Deviance					$\rho_{pc} = -.42^d (k = 8; N = 2,934)$
CWBs—Absenteeism					$\rho_{rc} = -.26^{qq} (k = 13; N = 6,276)$ $\rho_{rc} = -.06^{qq} (k = 10; N = 2,155)$
Procrastination	$\rho = -.73^{ss} (k = 21; N = 3,840)$	$\rho = -.45^{ss} (k = 25; N = 4,757)$		$\rho = -.43^{ss} (k = 34; N = 6,171)$	$\rho = -.75^{ss} (k = 20; N = 4,012)$

(continued overleaf)

TABLE 9.1 (continued)

Work Outcome (Criterion Construct)	Conscientiousness Facet			Global Conscientiousness
	Cautiousness/Impulse Control	Order	Dependability	Achievement
Workplace Safety (Accidents/Injuries—reversed)				$\rho_{pc} = .26^h$ ($k = 4$; $N = 852$) $\rho_{rc} = .06^{qq}$ ($k = 6$; $N = 2,094$)
Goal Setting				$\rho_{pc} = .28^x$ ($k = 13$; $N = ?$) $\rho_{pc} = .28^x$ ($k = 18$; $N = 2,211$)
Effort		$r = .14^p$ ($k = 11$; $N = 25,408$)	$r = .21^p$ ($k = 4$; $N = 15,530$)	
Getting Ahead		$\rho = .20^o$ ($k = 42$; $N = 5,017$)	$\rho = .26^o$ ($k = 42$; $N = 5,017$)	
Getting Along/Teamwork/Interpersonal Effectiveness		$\rho = .31^o$ ($k = 26$; $N = 2,949$) $r = .17^p$ ($k = 25$; $N = 1,340$)	$\rho = .17^o$ ($k = 26$; $N = 2,949$) $r = .14^p$ ($k = 3$; $N = 233$)	$\rho = .20^{cc}$ ($k = 10$; $N = 1,491$)
Innovation & Creativity		$r = -.07^p$ ($k = 5$; $N = 268$)	$r = .14^p$ ($k = 2$; $N = 116$)	$r = .07^m$ ($k = 48$; scientists) $r = -.29^m$ ($k = 52$; artists) $\rho_{pc} = .00^{xx}$ ($k = 3$; $N = ?$) $\rho_{pc} = -.06^{yy}$ ($k = 5$; $N = 946$; field) $\rho_{pc} = .13^{yy}$ ($k = 3$; $N = 707$; lab)
Leadership & Leadership Effectiveness		$\rho_{pc} = .30^y$ ($k = 16$; $N = 5,020$)	$\rho_{pc} = .35^y$ ($k = 16$; $N = 4,625$)	$\rho_{pc} = .28^k$ ($k = 39$; $N = 10,056$) $\rho_{pc} = .28^v$ ($k = 35$; $N = 7,510$) $\rho_{pc} = .05^v$ (business; $k = 8$; $N = ?$) $\rho_{pc} = .17^v$ (gov/military; $k = 6$; $N = ?$) $\rho_{pc} = .36^v$ (students; $k = 21$; $N = ?$)

Leadership—Transformational						$\rho_{\text{pc}} = .13^{\text{e}}$ ($k = 18$; $N = 3,516$)
Leadership—Transactional						$\rho_{\text{pc}} = -.11^{\text{e}}$ ($k = 7$; $N = 1,564$)
Knowledge, Education, Task & Skill Criteria:						
Task Performance/ Proficiency	$\rho_{\text{pc}} = .11^{\text{l}}$ ($k = 18$; $N = 5,921$)	$\rho_{\text{pc}} = .16^{\text{l}}$ ($k = 26$; $N = 2,998$)	$r = .07^{\text{f}}$ ($k = 9$; $N = 4,362$) $\rho_{\text{pc}} = .17^{\text{l}}$ ($k = 11$; $N = 934$) $r = .05^{\text{p}}$ ($k = 13$; $N = 25,327$) $r = .13^{\text{f}}$ ($k = 9$; $N = 4,362$)	$r = .09^{\text{f}}$ ($k = 9$; $N = 4,362$) $\rho_{\text{pc}} = .25^{\text{l}}$ ($k = 26$; $N = 3,074$) $r = .02^{\text{p}}$ ($k = 6$; $N = 15,554$) $r = .09^{\text{f}}$ ($k = 9$; $N = 4,362$)	$\rho = .16^{\text{s}}$ ($k = 12$; $N = 2,197$)	
Job Knowledge			$r = .12^{\text{p}}$ ($k = 42$; $N = 18,661$)	$r = .29^{\text{p}}$ ($k = 31$; $N = 12,639$)	$\rho_{\text{pc}} = .22^{\text{nn}}$ ($k = 138$; $N = 70,926$)	
Educational Outcomes			$r = .11^{\text{p}}$ ($k = 34$; $N = 4,710$)	$\rho_{\text{pc}} = .17^{\text{i}}$ ($k = 2$; $N = 356$) $r = .21^{\text{p}}$ ($k = 9$; $N = 1,160$)	$\rho = .23^{\text{b}}$ ($k = 17$; $N = 3,585$)	
Training Outcomes (e.g., skill acquisition, grades)					$\rho_{\text{pc}} = -.05^{\text{i}}$ ($k = 6$; $N = 839$) $\rho = .03^{\text{s}}$ ($k = 3$; $N = 741$) $\rho = .24^{\text{pp}}$ ($k = 3$; $N = 324$; Europe)	
Job & Career Satisfaction					$\rho_{\text{pc}} = .26^{\text{w}}$ ($k = 79$; $N = 21,719$) $\rho_{\text{pc}} = .14^{\text{ee}}$ ($k = 6$; $N = 10,566$)	
Engagement/Organizational Commitment					$\rho_{\text{pc}} = .44^{\text{i}}$ ($k = 2$; $N = 364$)	$\rho_{\text{pc}} = .38^{\text{i}}$ ($k = 2$; $N = 417$)
Objective Criteria:						
Awards			$r = .07^{\text{f}}$ ($k = 9$; $N = 4,362$)	$r = .20^{\text{f}}$ ($k = 9$; $N = 4,362$)		
Tenure (Turnover—reversed)						$\rho = .12^{\text{b}}$ ($k = 19$; $N = 2,759$) $\rho_{\text{rc}} = .31^{\text{qq}}$ ($k = 5$; $N = 748$) $\rho = .22^{\text{ww}}$ ($k = 17$; $N = 1,631$)

(continued overleaf)

TABLE 9.1 (continued)

Work Outcome (Criterion Construct)	Conscientiousness Facet				Global Conscientiousness
	Cautiousness/Impulse Control	Order	Dependability	Achievement	
Salary					$\rho = .17^b$ ($k = 5$; $N = 718$) $\rho_{pc} = .07^{ee}$ ($k = 6$; $N = 6,286$)
Level (authority; promotion)			$\rho = -.14^r$ ($k = 10$; $N = 1,065$)	$\rho = .19^r$ ($k = 27$; $N = 4,100$)	$\rho = .05^r$ ($k = 44$; $N = 6,667$)
Miscellaneous:					
First Impression					$\rho_{pc} = .27^j$ ($k = 7$; $N = 1,013$)

Note: The following footnotes are for Tables 1-6.

r = sample-size weighted mean observed correlation (uncorrected validity)

ρ_p = corrected for unreliability in predictor

ρ_c = corrected for unreliability in criterion

ρ_{rc} = corrected for range restriction and unreliability in criterion

ρ_{pc} = corrected for unreliability in predictor and criterion

ρ = corrected for range restriction and unreliability in predictor and criterion

^aArthur, Barrett, and Alexander (1991). ^bBarrick and Mount (1991). ^cBarrick and Zimmerman (2005). ^dBerry, Ones, and Sackett (2007). ^eBono and Judge (2004). ^fBorman, White, Pulakos, and Oppler (1991); see also Hough, Eaton, Dunnette, Kamp, and McCloy (1990), as well as McHenry, Hough, Toquam, Hanson, and Ashworth (1990). ^gBorman, Penner, Allen, and Motowidlo (2001). ^hChristian, Bradley, Wallace, and Burke (2009). ⁱColquitt, LePine, and Noe (2000). ^jConnelly and Ones (2010). ^kDerue, Nahrgang, Wellman, and Humphrey (2011). ^lDudley, Orvis, Lebiecki, and Cortina (2006). ^mFeist (1998). ⁿFrei and McDaniel (1998). ^oHogan and Holland (2003). ^pHough (1992). ^qHough and Dilchert (2007). ^rHough, Ones, and Viswesvaran (1998). ^sHurtz and Donovan (2000). ^tJoseph and Newman (2010). ^uJudge and Bono (2001). ^vJudge, Bono, Ilies, and Gerhardt (2002). ^wJudge, Heller, and Mount (2002). ^xJudge and Ilies (2002). ^yKish-Gephart, Harrison, and Treviño (2010). ^zLePine, Erez, and Johnson (2002). ^{aa}McCune, Cadiz, Drown, and Bodner (2007). ^{bb}Mol, Born, Willemsen, and Van Der Molen (2005). ^{cc}Mount, Barrick, and Stewart (1998). ^{dd}Ny, Sorensen, and Eby (2006). ^{ee}Ny, Eby, Sorensen, and Feldman (2005). ^{ff}Ones, Hough, and Viswesvaran (1998). ^{gg}Ones and Viswesvaran (2008). ^{hh}Ones and Viswesvaran (2001). ⁱⁱOnes, Viswesvaran, and Schmidt (1993). ^{jj}Ones, Viswesvaran, and Schmidt (2003). ^{kk}Ones, Dilchert, Viswesvaran, and Judge (2007); letter(s) after the ^{kk} is the code for the article from which the meta-analytic predictor-criterion correlations were obtained. For example, ^{kk/g} indicates that the original meta-analytic data were from Borman, Penner, Allen, and Motowidlo (2001), the article that was coded ^g as listed above. ^{ll}Organ and Ryan (1995). ^{mm}Peeters, Van Tuijl, Rutte, and Reymen (2006). ⁿⁿPoropat (2009). ^{oo}Rauch and Frese (2007). ^{pp}Salgado (1997). ^{qq}Salgado (2002). ^{rr}Salgado (2003). ^{ss}Steel (2007). ^{tt}Thoresen, Kaplan, Barsky, deChermont, and Warren (2003). ^{uu}Vinchor, Schippmann, Switzer, and Roth (1998). ^{vv}Wang, Bowling, & Eschleman (2009). ^{ww}Zimmerman (2008). ^{xx}Eder and Sawyer (2007). ^{yy}Harrison, Neff, Schwall, and Zhao (2006). ^{zz}Barrick, Mount, and Judge (2001).

TABLE 9.2 Meta-analytic Criterion-Related Validities of Self-Report Emotional Stability and Its Facets for Predicting Work Outcomes

Work Outcome (Criterion Construct)	Emotional Stability Facet		Global Emotional Stability
	Self-Esteem	Low Anxiety	
Overall (Individual) Job Performance	$\rho = .26^u$ ($k = 40$; $N = 5,145$)		$\rho = .07^b$ ($k = 87$; $N = 11,635$) $r = .11^p$ ($k = 186$; $N = 28,587$) $\rho = .15^s$ ($k = 35$; $N = 5,027$) $\rho = .19^u$ ($k = 20$; $N = 4,106$) $\rho = .12^{pp}$ ($k = 22$; $N = 2,799$; Europe) $\rho_{rc} = .16^{rr}$ ($k = 72$; $N = 10,786$; FFM scales) $\rho_{pc} = .14^{mm}$ ($k = 4$; $N = 236$; professional teams) $\rho_{pc} = -.04^{mm}$ ($k = 5$; $N = 261$; student teams)
Overall (Team) Performance			
Specific Occupation Success:			
Sales Effectiveness			$\rho = .07^b$ ($k = 19$; $N = 2,486$) $r = .18^p$ ($k = 3$; $N = 778$) $\rho = .15^s$ ($k = 7$; $N = 799$) $\rho_{rc} = .10^{uu}$ ($k = 24$; $N = 3,134$; ratings) $\rho_{rc} = -.12^{uu}$ ($k = 14$; $N = 2,157$; objective sales)
Skilled/Semiskilled			$\rho = .12^b$ ($k = 26$; $N = 3,694$) $\rho = .09^s$ ($k = 11$; $N = 1,874$)
Customer Service			$\rho = .13^s$ ($k = 10$; $N = 1,614$) $\rho = .12^{cc}$ ($k = 7$; $N = 908$)
Managerial Effectiveness			$\rho = .08^b$ ($k = 55$; $N = 10,324$) $\rho = .08^r$ ($k = 202$; $N = 69,889$) $\rho = .13^s$ ($k = 4$; $N = 495$) $\rho_{pc} = .10^{bb}$ ($k = 12$; $N = 1,189$) $r = .19^p$ ($k = 13$; $N = 3,880$)
Expatriate Effectiveness			
Combat Effectiveness			
Work Success Components:			
Contextual/Org. Citizenship (OCBs)—Overall			$r = .16^g$ ($k = 5$; $N = 970$)
OCBs—Dedication			$\rho = .14^s$ ($k = 15$; $N = 2,581$)
OCBs—Interpersonal Facilitation			$\rho = .17^s$ ($k = 21$; $N = 3,685$)
OCBs—General Compliance			$\rho_{pc} = .12^{ll}$ ($k = 5$; $N = 847$)
OCBs—Altruism			$\rho_{pc} = .06^{ll}$ ($k = 6$; $N = 1,201$)
Counterproductive Work Behavior (CWBs ¹)—Overall			$\rho_{pc} = -.26^d$ (estimated) $r = -.15^p$ ($k = 9$; $N = 21,431$) $\rho_{pc} = -.24^d$ ($k = 10$; $N = 2,842$)
CWBs ¹ —Interpersonal Deviance			$\rho_{pc} = -.23^d$ ($k = 7$; $N = 2,300$)
CWBs ¹ —Organizational Deviance			$\rho_{rc} = -.06^{qq}$ ($k = 15$; $N = 3,107$)
CWBs ¹ —Absenteeism			$\rho_{rc} = .04^{qq}$ ($k = 12$; $N = 2,491$)
Procrastination	$\rho = -.32^{ss}$ ($k = 33$; $N = 5,748$)		$\rho = -.28^{ss}$ ($k = 59$; $N = 10,720$)
Workplace Safety (Accidents/Injuries—rev.)			$\rho_{pc} = .19^h$ ($k = 12$; $N = 5,129$) $\rho_{rc} = .08^{qq}$ ($k = 5$; $N = 2,121$)
Goal Setting	$\rho_{pc} = .27^x$ ($k = 7$; $N = ?$)		$\rho_{pc} = .29^x$ ($k = 19$; $N = 2,780$)

(continued overleaf)

TABLE 9.2 (continued)

Work Outcome (Criterion Construct)	Emotional Stability Facet		Global Emotional Stability
	Self-Esteem	Low Anxiety	
Effort			$r = .16^p$ ($k = 15$; $N = 9,562$)
Getting Ahead			$\rho = .22^o$ ($k = 42$; $N = 5,017$)
Getting Along/Teamwork/ Interpersonal Effectiveness			$\rho = .34^o$ ($k = 26$; $N = 2,949$) $r = .13^p$ ($k = 31$; $N = 2,067$) $\rho = .19^{cc}$ ($k = 10$; $N = 1,491$)
Innovation & Creativity			$r = .02^m$ ($k = 66$; scientists) $r = -.07^m$ ($k = 128$; artists) $r = -.05^p$ ($k = 8$; $N = 442$) $\rho_{pc} = .04^{yy}$ ($k = 3$; $N = 448$; field) $\rho_{pc} = -.03^{yy}$ ($k = 4$; $N = 1,332$; lab)
Leadership & Leadership Effectiveness	$\rho_{pc} = .19^v$ ($k = 9$; $N = 7,451$)		$\rho_{pc} = .24^k$ ($k = 51$; $N = 8,960$) $\rho_{pc} = .24^v$ ($k = 48$; $N = 8,025$) $\rho_{pc} = .15^v$ (business; $k = 9$; $N = ?$) $\rho_{pc} = .23^v$ (gov/military; $k = 12$; $N = ?$) $\rho_{pc} = .27^v$ (students; $k = 27$; $N = ?$)
Leadership—Transformational			$\rho_{pc} = .17^e$ ($k = 18$; $N = 3,380$)
Leadership—Transactional			$\rho_{pc} = -.05^e$ ($k = 8$; $N = 1,627$)
Knowledge, Education, Task, & Skill Criteria:			
Task Performance/Proficiency			$r = .05^p$ ($k = 23$; $N = 9,364$) $\rho = .14^s$ ($k = 8$; $N = 1,243$)
Educational Outcomes			$r = .20^p$ ($k = 162$; $N = 70,588$) $\rho_{pc} = .02^{nn}$ ($k = 114$; $N = 59,554$)
Training Outcomes (e.g., skill acquisition, grades)		$\rho_{pc} = .15^i$ ($k = 4$; $N = 368$)	$\rho = .07^b$ ($k = 19$; $N = 3,283$) $r = .12^p$ ($k = 69$; $N = 8,685$) $\rho = .09^s$ ($k = 2$; $N = 644$) $\rho = .18^{pp}$ ($k = 6$; $N = 470$; Europe)
Job & Career Satisfaction	$\rho = .26^u$ ($k = 56$; $N = 20,819$)		$\rho = .24^u$ ($k = 21$; $N = 7,658$) $\rho_{pc} = .29^w$ ($k = 92$; $N = 24,527$) $\rho_{pc} = .36^{ee}$ ($k = 6$; $N = 10,566$) $\rho_{pc} = .28^{tt}$ ($k = 42$; $N = 13,500$) $\rho_{pc} = .23^{tt}$ ($k = 3$; $N = 713$)
Engagement/Organizational Commitment			
Objective Criteria:			
Tenure (Turnover—reversed)	$r = .17^c$ ($k = 2$; $N = 445$)		$\rho = .02^b$ ($k = 13$; $N = 1,495$) $\rho_{rc} = .35^{qq}$ ($k = 4$; $N = 554$) $\rho = .20^{ww}$ ($k = 19$; $N = 1,824$)
Salary			$\rho = -.01^b$ ($k = 4$; $N = 666$) $\rho_{pc} = .12^{ee}$ ($k = 7$; $N = 6,433$)
Level (authority; promotion)			$\rho = .25^r$ ($k = 30$; $N = 6,772$)
Miscellaneous:			
First Impression			$\rho_{pc} = .20^j$ ($k = 7$; $N = 1,013$)

See Table 9.1 footnotes.

TABLE 9.3 Meta-analytic Self-Report Criterion-Related Validities of Agreeableness for Predicting Work Outcomes

Work Outcome (Criterion Construct)	Global Agreeableness
Overall (Individual) Job Performance	$\rho = .06^b$ ($k = 80$; $N = 11,526$) $r = .04^p$ ($k = 69$; $N = 12,722$) $\rho = .12^s$ ($k = 38$; $N = 5,803$) $\rho = -.01^{pp}$ ($k = 19$; $N = 2,574$; Europe) $\rho_{rc} = .13^{rr}$ ($k = 68$; $N = 10,716$; FFM scales) $\rho_{pc} = .20^k$ ($k = 2$; $N = 84$) $\rho_{pc} = .51^{mm}$ ($k = 4$; $N = 236$; professional teams) $\rho_{pc} = .02^{mm}$ ($k = 5$; $N = 261$; student teams)
Overall (Team) Performance	
Specific Occupation Success:	
Sales Effectiveness	$\rho = .00^b$ ($k = 16$; $N = 2,344$) $\rho = .06^s$ ($k = 8$; $N = 959$) $\rho_{rc} = .06^{uu}$ ($k = 23$; $N = 2,342$; ratings) $\rho_{rc} = -.03^{uu}$ ($k = 12$; $N = 918$; objective sales)
Skilled/Semiskilled	$\rho = .06^b$ ($k = 28$; $N = 4,585$) $\rho = .11^s$ ($k = 12$; $N = 2,385$)
Customer Service	$\rho = .19^s$ ($k = 11$; $N = 1,719$) $\rho = .13^{cc}$ ($k = 7$; $N = 908$)
Managerial Effectiveness	$\rho = .10^b$ ($k = 47$; $N = 8,597$) $\rho = .04^r$ ($k = 99$; $N = 42,218$) $\rho = -.04^s$ ($k = 4$; $N = 495$) $\rho_{pc} = .11^{bb}$ ($k = 11$; $N = 1,021$)
Expatriate Effectiveness	
Components of Work Success:	
Contextual/Organizational Citizenship (OCBs)—Overall	$r = .13^g$ ($k = 7$; $N = 1,554$)
OCBs—Dedication	$\rho = .10^s$ ($k = 17$; $N = 3,197$)
OCBs—Interpersonal Facilitation	$\rho = .20^s$ ($k = 23$; $N = 4,301$)
OCBs—General Compliance	$\rho_{pc} = .11^{ll}$ ($k = 6$; $N = 916$)
OCBs—Altruism	$\rho_{pc} = .13^{ll}$ ($k = 6$; $N = 916$)
Counterproductive Work Behavior (CWBs ¹)—Overall	$\rho_{pc} = -.44^d$ (estimated) $r = -.08^p$ ($k = 4$; $N = 24,259$) $\rho_{pc} = -.46^d$ ($k = 10$; $N = 3,336$) $\rho_{pc} = -.32^d$ ($k = 8$; $N = 2,934$) $\rho_{rc} = -.20^{qq}$ ($k = 9$; $N = 1,299$) $\rho_{rc} = .04^{qq}$ ($k = 8$; $N = 1,339$) $\rho = -.14^{ss}$ ($k = 24$; $N = 5,001$) $\rho_{rc} = .01^{qq}$ ($k = 4$; $N = 1,540$) $\rho_{pc} = -.29^x$ ($k = 4$; $N = 373$) $\rho = .11^o$ ($k = 42$; $N = 5,017$) $\rho = .23^o$ ($k = 26$; $N = 2,949$) $r = .17^p$ ($k = 7$; $N = 329$) $\rho = .27^{cc}$ ($k = 10$; $N = 1,491$)
CWBs ¹ —Interpersonal Deviance.	
CWBs ¹ —Organizational Deviance	
CWBs ¹ —Absenteeism	
Procrastination	
Workplace Safety (Accidents/Injuries—rev.)	
Goal Setting	
Getting Ahead	
Getting Along/Teamwork/ Interpersonal Effectiveness	
Innovation & Creativity	$r = -.03^m$ ($k = 64$; scientists) $r = -.10^m$ ($k = 63$; artists) $r = -.29^p$ ($k = 3$; $N = 174$) $\rho_{pc} = -.04^{yy}$ ($k = 3$; $N = 448$; field) $\rho_{pc} = .08^{yy}$ ($k = 3$; $N = 707$; lab) $\rho_{pc} = .08^k$ ($k = 45$; $N = 10,507$) $\rho_{pc} = .08^v$ ($k = 42$; $N = 9,801$) $\rho_{pc} = -.04^v$ (business; $k = 10$; $N = ?$) $\rho_{pc} = -.04^v$ (gov/military; $k = 11$; $N = ?$) $\rho_{pc} = .18^v$ (students; $k = 21$; $N = ?$) $\rho_{pc} = .14^e$ ($k = 20$; $N = 3,916$) $\rho_{pc} = -.12^e$ ($k = 7$; $N = 1,564$)
Leadership & Leadership Effectiveness	
Leadership—Transformational	
Leadership—Transactional	

(continued overleaf)

TABLE 9.3 (continued)

Work Outcome (Criterion Construct)	Global Agreeableness
Knowledge, Education, Task, & Skill Criteria:	
Task Performance/ Proficiency	$r = .02^p$ ($k = 4$; $N = 7,837$) $\rho = .08^s$ ($k = 9$; $N = 1,754$)
Educational Outcomes	$r = .01^p$ ($k = 15$; $N = 7,330$) $\rho_{pc} = .07^{nn}$ ($k = 109$; $N = 58,522$)
Training Outcomes (e.g., skill acquisition, grades)	$\rho = .10^b$ ($k = 19$; $N = 3,685$) $r = .08^p$ ($k = 7$; $N = 988$) $\rho = .21^s$ ($k = 2$; $N = 644$) $\rho = .19^{pp}$ ($k = 5$; $N = 415$; Europe) $\rho_{pc} = .17^w$ ($k = 38$; $N = 11,856$) $\rho_{pc} = .11^{ee}$ ($k = 5$; $N = 4,634$)
Job & Career Satisfaction	
Objective Criteria:	
Tenure (Turnover—reversed)	$\rho = .09^b$ ($k = 15$; $N = 1,838$) $\rho_{rc} = .22^{qq}$ ($k = 4$; $N = 554$) $\rho = .27^{ww}$ ($k = 15$; $N = 1,532$)
Salary	$\rho = -.02^b$ ($k = 2$; $N = 121$) $\rho_{pc} = -.10^{ee}$ ($k = 6$; $N = 6,286$)
Level (authority; promotion)	$\rho = .12^r$ ($k = 16$; $N = 2,904$)
Miscellaneous:	
First Impression	$\rho_{pc} = .26^j$ ($k = 7$; $N = 1,013$)

See Table 9.1 footnotes.

One approach to developing the database is to conduct primary studies that report relationships between predictor constructs and job components and then use meta-analysis to cumulate the results of those studies. This is a practical strategy because thousands of criterion-related validation studies have been conducted that could potentially serve as input to the database, and future validation studies can be designed with contributing to the database in mind (Johnson et al., 2010). To effectively build a database using this strategy, standard taxonomies of predictor and criterion constructs must be agreed upon. We reviewed some more specific personality taxonomies earlier in this chapter. Although there is little agreement on the appropriate personality taxonomy to use in researching personality–performance relationships, there have at least been concerted efforts on the part of recent researchers to organize personality variables into a taxonomy that makes sense.

On the criterion side, we are not nearly as far along. As evidenced by Tables 9.1 through 9.6, most meta-analyses of personality–performance relationships have been limited to whatever criteria happen to be available, and these criteria do not come close to representing the entire domain of individual job performance. Recent work on criterion taxonomies has advanced our understanding of the individual job performance domain.

At the broad level, both Campbell (1990) and Bartram (2005) have proposed eight categories that can be used to

define virtually any job. Campbell's model consists of (a) job-specific task proficiency, (b) non-job-specific task proficiency, (c) written and oral communication proficiency, (d) demonstrate effort, (e) maintain personal discipline, (f) facilitate peer and team performance, (g) supervision/leadership, and (h) management/administration. Bartram's "Great Eight Competencies" are (a) leading and deciding, (b) supporting and cooperating, (c) interacting and presenting, (d) analyzing and interpreting, (e) creating and conceptualizing, (f) organizing and executing, (g) adapting and coping, and (h) enterprising and performing.

O*NET represents task behaviors with 42 Generalized Work Activities (Jeanneret, Borman, Kubisiak, & Hansen, 1999), which are categorized into nine broader categories: (a) looking for and receiving job-related information; (b) identifying and evaluating job-relevant information; (c) information and data processing; (d) reasoning and decision making; (e) performing physical and manual work activities; (f) performing complex and technical activities; (g) communicating and interacting; (h) coordinating, developing, managing, and advising; and (i) administering.

Johnson (2003) proposed a multilevel taxonomy of performance dimensions to be used specifically for building a synthetic validity database. This taxonomy has three components at the highest level: (a) task performance, (b) citizenship performance, and (c) adaptive performance. Level 2 defined these components with five dimensions from

TABLE 9.4 Meta-analytic Criterion-Related Validities of Self-Report Extraversion and Its Facets for Predicting Work Outcomes

Work Outcome (Criterion Construct)	Extraversion Facet			Global Extraversion
	Dominance	Sociability	Activity/Energy Level	
Overall (Individual) Job Performance	$r = .09^p$ ($k = 248$; $N = 30,642$)	$r = .02^p$ ($k = 31$; $N = 3,782$)		$\rho = .10^b$ ($k = 89$; $N = 12,396$) $\rho = .09^s$ ($k = 37$; $N = 5,809$) $\rho = .09^{pp}$ ($k = 22$; $N = 2,799$; Europe) $\rho_{rc} = .07^{rr}$ ($k = 75$; $N = 11,940$; FFM scales) $\rho_{pc} = .00^k$ ($k = 3$; $N = 135$) $\rho_{pc} = .15^{mm}$ ($k = 4$; $N = 236$; professional teams) $\rho_{pc} = -.05^{mm}$ ($k = 5$; $N = 261$; student teams)
Overall (Team) Performance				
Specific Occupation Success:				
Sales Effectiveness	$r = .25^p$ ($k = 7$; $N = 1,111$) $\rho_{rc} = .28^{uu}$ ($k = 25$; $N = 2,907$; ratings) $\rho_{rc} = .26^{uu}$ ($k = 14$; $N = 2,278$; objective sales)	$\rho_{rc} = .12^{uu}$ ($k = 18$; $N = 2,389$; ratings) $\rho_{rc} = .15^{uu}$ ($k = 4$; $N = 279$; objective sales)		$\rho = .15^b$ ($k = 22$; $N = 2,316$) $\rho = .16^s$ ($k = 8$; $N = 1,044$) $\rho_{rc} = .018^{uu}$ ($k = 27$; $N = 3,112$; ratings) $\rho_{rc} = .22^{uu}$ ($k = 18$; $N = 2,629$; objective sales)
Skilled/Semiskilled				$\rho = .01^b$ ($k = 23$; $N = 3,888$) $\rho = .01^s$ ($k = 12$; $N = 2,385$)
Customer Service				$\rho = .11^s$ ($k = 10$; $N = 1,640$) $\rho = .07^{cc}$ ($k = 6$; $N = 829$)
Managerial Effectiveness	$\rho = .27^r$ ($k = 125$; $N = 11,823$)	$\rho = -.02^r$ ($k = 102$; $N = 19,454$)	$\rho = .20^r$ ($k = 22$; $N = 8,937$)	$\rho = .18^b$ ($k = 59$; $N = 11,335$) $\rho = .09^r$ ($k = 379$; $N = 108,607$) $\rho = .13^s$ ($k = 4$; $N = 495$)
Expatriate Effectiveness				$\rho_{pc} = .17^{bb}$ ($k = 12$; $N = 1,114$)
Combat Effectiveness	$r = .08^p$ ($k = 9$; $N = 2,695$)	$r = -.02^p$ ($k = 2$; $N = 600$)		
Work Success Components:				
Contextual/ Organizational Citizenship (OCBs)—Overall				$r = .06^g$ ($k = 7$; $N = 1,728$)
OCBs—Dedication				$\rho = .05^s$ ($k = 16$; $N = 3,130$)
OCBs—Interpersonal Facilitation				$\rho = .11^s$ ($k = 21$; $N = 4,155$)
OCBs—General Compliance				$\rho_{pc} = .07^{ll}$ ($k = 6$; $N = 934$)
OCBs—Altruism				$\rho_{pc} = .08^{ll}$ ($k = 5$; $N = 869$)

(continued overleaf)

TABLE 9.4 (continued)

Work Outcome (Criterion Construct)	Extraversion Facet			Global Extraversion
	Dominance	Sociability	Activity/Energy Level	
Counterproductive Work Behavior (CWBs ¹)— Overall	$r = -.06^p$ ($k = 14$; $N = 38,578$)			$\rho_{pc} = -.03^d$ (estimated) ρ
CWBs ¹ —Interpersonal Deviance				$\rho_{pc} = .02^d$ ($k = 8$; $N = 2,360$)
CWBs ¹ —Organizational Deviance				$\rho_{pc} = -.09^d$ ($k = 5$; $N = 1,836$)
				$\rho_{rc} = .01^{qq}$ ($k = 12$; $N = 2,383$)
CWBs ¹ —Absenteeism				$\rho_{rc} = .08^{qq}$ ($k = 10$; $N = 1,799$)
Procrastination				$\rho = -.14^{ss}$ ($k = 27$; $N = 5,032$)
Workplace Safety (Accidents/Injuries—reversed)				$\rho_{pc} = .07^h$ ($k = 5$; $N = 2,083$)
				$\rho_{rc} = .04^{qq}$ ($k = 7$; $N = 2,341$)
Goal Setting Effort	$r = .17^p$ ($k = 16$; $N = 17,156$)			$\rho_{pc} = .15^x$ ($k = 5$; $N = 498$)
Getting Ahead		$\rho = .04^o$ ($k = 42$; $N = 5,017$)		
Getting Along/Teamwork/Interpersonal Effectiveness	$r = .08^p$ ($k = 39$; $N = 2,307$)	$\rho = .01^o$ ($k = 26$; $N = 2,949$)		$\rho = .14^{cc}$ ($k = 9$; $N = 1,412$)
Innovation & Creativity	$r = .19^m$ ($k = 42$; scientists) $r = .08^m$ ($k = 42$; artists) $r = .21^p$ ($k = 11$; $N = 550$)	$r = .07^m$ ($k = 23$; scientists) $r = .01^m$ ($k = 35$; artists) $r = -.25^p$ ($k = 2$; $N = 116$)		$r = .14^m$ ($k = 135$; scientists) $r = .08^m$ ($k = 148$; artists) $\rho_{pc} = .04^{yy}$ ($k = 3$; $N = 448$; field) $\rho_{pc} = .03^{yy}$ ($k = 4$; $N = 1,332$; lab)
Leadership & Leadership Effectiveness	$\rho_{pc} = .37^v$ ($k = 31$; $N = 7,692$)	$\rho_{pc} = .37^v$ ($k = 19$; $N = 5,827$)		$\rho_{pc} = .31^k$ ($k = 63$; $N = 12,640$) $\rho_{pc} = .31^v$ ($k = 60$; $N = 11,705$) $\rho_{pc} = .25^v$ (business; $k = 13$; $N = ?$) $\rho_{pc} = .16^v$ (gov/military; $k = 10$; $N = ?$) $\rho_{pc} = .40^v$ (students; $k = 37$; $N = ?$)
Leadership—Transformational				$\rho_{pc} = .24^e$ ($k = 20$; $N = 3,692$)
Leadership—Transactional				$\rho_{pc} = -.09^e$ ($k = 6$; $N = 1,310$)
Knowledge, Education, Task, & Skill Criteria:				
Task Performance/Proficiency	$r = .02^p$ ($k = 23$; $N = 17,001$)	$r = .06^p$ ($k = 2$; $N = 736$)		$\rho = .07^s$ ($k = 9$; $N = 1,839$)

TABLE 9.4 (continued)

Work Outcome (Criterion Construct)	Extraversion Facet			Global Extraversion
	Dominance	Sociability	Activity/Energy Level	
Educational Outcomes	$r = .12^p$ ($k = 128$; $N = 63,057$)	$r = .01^p$ ($k = 9$; $N = 2,953$)		$\rho_{pc} = -.01^{nn}$ ($k = 113$; $N = 59,986$)
Training Outcomes (e.g., skill acquisition, grades)	$r = .07^p$ ($k = 70$; $N = 8,389$)			$\rho = .26^b$ ($k = 17$; $N = 3,101$) $\rho = .19^s$ ($k = 2$; $N = 644$) $\rho = .02^{pp}$ ($k = 4$; $N = 383$; Europe)
Job & Career Satisfaction				$\rho_{pc} = .25^w$ ($k = 75$; $N = 20,184$) $\rho_{pc} = .27^{ee}$ ($k = 6$; $N = 10,566$) $\rho_{pc} = .22^{tt}$ ($k = 37$; $N = 12,023$) $\rho_{pc} = .22^{tt}$ ($k = 2$; $N = 492$)
Engagement/Org. Commitment				
Objective Criteria:				
Tenure (Turnover—reversed)				$\rho = -.03^b$ ($k = 13$; $N = 1,437$) $\rho_{rc} = .20^{qq}$ ($k = 4$; $N = 554$) $\rho = .04^{ww}$ ($k = 18$; $N = 1,608$)
Salary				$\rho = .06^b$ ($k = 4$; $N = 666$) $\rho_{pc} = .10^{ee}$ ($k = 7$; $N = 6,610$)
Level (authority; promotion)	$\rho = .24^r$ ($k = 26$; $N = 3,536$)	$\rho = .14^r$ ($k = 20$; $N = 4,437$)	$\rho = .42^r$ ($k = 10$; $N = 1,212$)	$\rho = .17^r$ ($k = 65$; $N = 10,586$)
Miscellaneous:				
First Impression				$\rho_{pc} = .37^j$ ($k = 7$; $N = 1,013$)

See Table 9.1 footnotes.

Campbell (1990); three citizenship performance dimensions from Borman et al. (2001); and two adaptive performance dimensions from Pulakos, Arad, Donovan, and Plamondon (2000). At Level 3, each Level 2 dimension is defined by more specific labels or descriptions from Pulakos et al. (2000), Borman and Brush (1993), and Campbell (1990). Johnson recommended that Level 2 dimensions be used for cumulating results across studies for meta-analyses, at least until sufficient data have been gathered to conduct meta-analyses at Level 3. This taxonomy may be a useful starting point for identifying the job components to be included in a synthetic validity database.

ETHNIC, GENDER, AND AGE GROUP SIMILARITIES AND DIFFERENCES

Most organizations desire a workforce that is representative of the community they serve, including characteristics

such as ethnic background, gender, and age. In the United States, minority groups, women, and people 40 years of age and older are legally protected classes and personnel decisions must be job-related to overcome legal challenges of adverse impact against a protected class. In such legal challenges, adverse impact is calculated by dividing the selection ratio of the protected group by the selection ratio of the majority group, and in the United States the four-fifths rule is often used as an indicator of disparate or adverse impact. Thus, if the selection ratio of the African American group is 60% that of the White group, the organization may need to defend the job-relatedness of its selection practices.

Group mean score differences and the overall selection ratio are important determinants of adverse impact. Depending upon the overall selection ratio, even fairly small group mean score differences can produce adverse impact against a protected group. Importantly, personality variables typically show small, if any, mean score differences between ethnic groups. When such differences

TABLE 9.5 Meta-analytic Criterion-Related Validities of Self-Report Openness to Experience and Its Facets for Predicting Work Outcomes

Work Outcome (Criterion Construct)	Openness to Experience Facet		Global Openness to Experience
	School Success	Intellect/Culture	
Overall (Individual) Job Performance			$\rho = -.03^b$ ($k = 55$; $N = 9,454$) $r = .01^p$ ($k = 36$; $N = 10,888$) $\rho = .06^s$ ($k = 33$; $N = 4,881$) $\rho = .01^{pp}$ ($k = 11$; $N = 1,629$; Europe) $\rho_{rc} = .08^{rr}$ ($k = 48$; $N = 7,562$; FFM scales)
Overall (Team) Performance			$\rho_{pc} = .13^k$ ($k = 2$; $N = 117$) $\rho_{pc} = .19^{mm}$ ($k = 3$; $N = 185$; professional teams) $\rho_{pc} = -.12^{mm}$ ($k = 3$; $N = 177$; student teams)
Specific Occupation Success:			
Sales Effectiveness			$\rho = -.02^b$ ($k = 12$; $N = 1,566$) $\rho = .04^s$ ($k = 6$; $N = 732$) $\rho_{rc} = .11^{uu}$ ($k = 8$; $N = 804$; ratings) $\rho_{rc} = .06^{uu}$ ($k = 6$; $N = 951$; objective sales)
Skilled/Semiskilled			$\rho = .01^b$ ($k = 16$; $N = 3,219$) $\rho = -.02^s$ ($k = 11$; $N = 1,874$)
Customer Service			$\rho = .17^s$ ($k = 9$; $N = 1,535$) $\rho = .17^{cc}$ ($k = 6$; $N = 829$)
Managerial Effectiveness		$\rho = .06^r$ ($k = 17$; $N = 3,291$)	$\rho = .08^b$ ($k = 37$; $N = 7,611$) $\rho = .08^r$ ($k = 110$; $N = 46,614$) $\rho = -.03^s$ ($k = 4$; $N = 495$) $\rho_{pc} = .06^{bb}$ ($k = 11$; $N = 1,023$)
Expatriate Effectiveness			
Work Success Components:			
Contextual/Organizational Citizenship (OCBs)—Overall			
OCBs—Dedication			$\rho = .01^s$ ($k = 14$; $N = 2,514$)
OCBs—Interpersonal Facilitation			$\rho = .05^s$ ($k = 19$; $N = 3,539$)
Counterproductive Work Behavior (CWBs ¹)—Overall			$\rho_{pc} = -.08^d$ (estimated) $r = -.15^p$ ($k = 2$; $N = 1,414$)
CWBs ¹ —Interpersonal Deviance			$\rho_{pc} = -.09^d$ ($k = 8$; $N = 2,360$)
CWBs ¹ —Organizational Deviance			$\rho_{pc} = -.04^d$ ($k = 5$; $N = 1,772$) $\rho_{rc} = .14^{qq}$ ($k = 8$; $N = 1,421$) $\rho_{rc} = .00^{qq}$ ($k = 8$; $N = 1,339$)
CWBs ¹ —Absenteeism			$\rho = .04^{ss}$ ($k = 16$; $N = 3,612$)
Procrastination			$\rho_{rc} = -.09^{qq}$ ($k = 5$; $N = 1,660$)
Workplace Safety (Accidents/Injuries—reversed)			
Goal Setting			$\rho_{pc} = .18^x$ ($k = 4$; $N = 262$)
Getting Ahead	$\rho = .15^o$ ($k = 32$; $N = 4,211$)	$\rho = .12^o$ ($k = 42$; $N = 5,017$)	
Getting Along/Teamwork/Interpersonal Effectiveness	$\rho = .12^o$ ($k = 22$; $N = 2,553$)	$\rho = .03^o$ ($k = 26$; $N = 2,949$)	$\rho = .10^{cc}$ ($k = 9$; $N = 1,412$)
Innovation & Creativity			$r = .18^m$ ($k = 52$; scientists) $r = .21^m$ ($k = 93$; artists) $\rho_c = .17^{xx}$ ($k = 7$; $N = ?$) $\rho_{pc} = .29^{yy}$ ($k = 4$; $N = 597$; field) $\rho_{pc} = .33^{yy}$ ($k = 3$; $N = 707$; lab)

TABLE 9.5 (continued)

Work Outcome (Criterion Construct)	Openness to Experience Facet		Global Openness to Experience
	School Success	Intellect/Culture	
Leadership & Leadership Effectiveness			$\rho_{pc} = .24^k$ ($k = 39$; $N = 7,762$) $\rho_{pc} = .23^v$ (business; $k = 9$; $N = ?$) $\rho_{pc} = .06^v$ (gov/military; $k = 6$; $N = ?$) $\rho_{pc} = .28^v$ (students; $k = 22$; $N = ?$)
Leadership—Transformational			$\rho_{pc} = .15^e$ ($k = 19$; $N = 3,887$)
Leadership—Transactional			$\rho_{pc} = .04^e$ ($k = 7$; $N = 1,564$)
Knowledge, Education, Task, & Skill			
Criteria:			
Task Performance/Proficiency			$r = .16^p$ ($k = 2$; $N = 700$)
Educational Outcomes			$r = .13^p$ ($k = 8$; $N = 3,628$) $\rho = -.01^s$ ($k = 7$; $N = 1,176$) $\rho_{pc} = .12^{nn}$ ($k = 113$; $N = 60,442$) $\rho = .25^b$ ($k = 14$; $N = 2,700$) $r = .02^p$ ($k = 35$; $N = 8,744$) $\rho = .14^s$ ($k = 2$; $N = 644$) $\rho = .17^{pp}$ ($k = 4$; $N = 477$; Europe) $\rho_{pc} = .02^w$ ($k = 50$; $N = 15,196$) $\rho_{pc} = .12^{ee}$ ($k = 7$; $N = 10,962$)
Training Outcomes (e.g., skill acquisition, grades)			
Job & Career Satisfaction			
Objective Criteria:			
Tenure (Turnover—reversed)			$\rho = -.11^b$ ($k = 12$; $N = 1,628$) $\rho_{rc} = .14^{qq}$ ($k = 4$; $N = 554$) $\rho = -.10^{ww}$ ($k = 16$; $N = 1,563$) $\rho = .05^b$ ($k = 2$; $N = 121$) $\rho_{pc} = .04^{ee}$ ($k = 7$; $N = 6,800$)
Salary			$\rho = .12^r$ ($k = 15$; $N = 2,985$)
Level (authority; promotion)		$\rho = .09^r$ ($k = 6$; $N = 1,416$)	
Miscellaneous:			
First Impression			$\rho_{pc} = .42^j$ ($k = 5$; $N = 989$)

See Table 9.1 footnotes.

do occur, the White mean score may be lower than the minority mean score. Also important, when such differences do occur, they may be at the facet level rather than at the broader level (e.g., FFM factor). It is the wise employer that understands known group score differences at both the facet and factor level. Such knowledge can be used to build selection systems that are likely to produce workforces that mirror the community.

There have been several quantitative summaries of group mean score differences between Whites and various ethnic groups, between men and women, and between people 40 years of age and older and people younger than 40 years of age (e.g., Duehr, Jackson, & Ones, 2003; Foldes et al., 2008; Hough et al., 2001). One comprehensive summary that examined ethnic, gender, and age differences found:

...some surprising results. Research clearly indicates that the setting, sample, the construct and the level of construct

specificity can all, either individually or in combination, moderate the magnitude of differences between groups. Employers using tests in employment settings need to assess accurately the requirements of work. When the exact nature of the work is specified, the appropriate predictors may or may not have adverse impact against some groups (Hough et al., 2001, p. 152).

One of the important findings from the Hough et al. (2001) summary is that comparisons between groups need to be made at both the facet and factor level. Comparisons made only at the factor level can obscure differences at the facet level, important differences that can be used to simultaneously increase criterion-related validity and decrease adverse impact.

Ethnic Group Similarities and Differences

Two quantitative summaries examined ethnic group mean score differences at both the factor and facet levels for

TABLE 9.6A Meta-analytic Criterion-Related Validities of Compound Personality Variables (Self-Report) for Predicting Work Outcomes

Work Outcome (Criterion Construct)	Compound Personality Variable					
	Personality-based Integrity	Customer Service	Managerial Potential	Locus of Control (Internal)	Emotional Intelligence	Autonomy
Overall (Individual) Job Performance	$\rho = .35^{ii}$ ($k = 138$; $N = 37,683$)	$\rho = .50^n$ ($k = 41$; $N = 6,945$) $\rho_c = .39^{gg}$ ($k = 33$; $N = 6,944$)		$r = .19^p$ ($k = 11$; $N = 2,517$) $\rho = .22^u$ ($k = 35$; $N = 4,310$) $\rho_{pc} = .13^{vv}$ ($k = 17$; $N = 4,005$)	$\rho = .18^t$ ($k = 10$; $N = 887$)	
Specific Occupation Success:						
Customer Service		$\rho_c = .34^{gg}$ ($k = 15$; $N = 4,401$)				
Managerial Effectiveness			$\rho = .42^{ff}$ ($k = 87$; $N = 11,009$)			
Entrepreneurship—Business Creation				$\rho_{pc} = .19^{oo}$ ($k = 24$; $N = 5,648$)		$\rho_{pc} = .31^{oo}$ ($k = 11$; $N = 4,256$)
Entrepreneurial Success				$\rho_{pc} = .13^{oo}$ ($k = 23$; $N = 3,959$)		$\rho_{pc} = .16^{oo}$ ($k = 8$; $N = 843$)
Work Success Components:						
Contextual/Organizational Citizenship (OCBs)—Overall			$\rho = .26^{ff}$ ($k = 16$; $N = 2,270$)	$r = .12^g$ ($k = 3$; $N = 599$)		
Counterproductive Work Behavior (CWBs ¹)—Overall	$\rho = -.29^{ii}$ ($k = 62$; $N = 93,092$; applicants, predictive)	$\rho_c = -.42^{gg}$ ($k = 5$; $N = 740$)	$\rho = -.30^{ff}$ ($k = 20$; $N = 2,948$)	$r = -.12^p$ ($k = 2$; $N = 8,333$) $\rho_{pc} = -.25^y$ ($k = 4$; $N = 713$)		
CWBs ¹ —Absenteeism	$\rho = -.36^{ij}$ ($k = 18$; $N = 5,435$) $\rho = -.33^{ij}$ ($k = 13$; $N = 4,922$; predictive)			$\rho_{pc} = -.13^{dd}$ ($k = 6$; $N = 2,927$) $\rho_{pc} = -.05^{vy}$ ($k = 2$; $N = 950$)		
Workplace Safety (Accidents/Injuries—reversed)				$r = .20^a$ ($k = 13$; $N = 1,909$) $\rho_{pc} = .26^h$ ($k = 4$; $N = 2,446$)		
Goal Setting				$\rho_{pc} = .30^x$ ($k = 8$; $N = ?$)		
Effort			$\rho = .13^{ff}$ ($k = 23$; $N = 23,672$)	$r = .13^p$ ($k = 6$; $N = 9,039$)		

Getting Along/Teamwork/ Interpersonal Effectiveness	$\rho = .14^{\text{ff}}$ ($k = 24$; $N = 26,405$)	
Leadership & Leadership Effectiveness	$\rho = .13^{\text{ff}}$ ($k = 16$; $N = 24,018$)	$\rho_{\text{pc}} = .13^{\text{v}}$ ($k = 15$; $N = 2,347$)
Knowledge, Education, Task, & Skill Criteria:		
Task Performance/Proficiency	$\rho = .24^{\text{ff}}$ ($k = 20$; $N = 5,123$)	$r = .06^{\text{p}}$ ($k = 2$; $N = 8,333$) $\rho_{\text{pc}} = .17^{\text{dd}}$ ($k = 36$; $N = 9,443$) $\rho_{\text{pc}} = .04^{\text{i}}$ ($k = 7$; $N = 386$) $r = .28^{\text{p}}$ ($k = 2$; $N = 225$) $\rho = .32^{\text{u}}$ ($k = 80$; $N = 18,491$) $\rho_{\text{pc}} = .33^{\text{dd}}$ ($k = 128$; $N = 30,622$) $\rho_{\text{pc}} = .47^{\text{ee}}$ ($k = 3$; $N = 668$) $\rho_{\text{pc}} = .27^{\text{vv}}$ ($k = 44$; $N = 12,197$) $\rho_{\text{pc}} = .42^{\text{vv}}$ ($k = 69$; $N = 16,010$; Work LC) $\rho_{\text{pc}} = .22^{\text{dd}}$ ($k = 34$; $N = 10,856$)
Training Outcomes (e.g., skill acquisition, grades)		
Job & Career Satisfaction		
Engagement/Organizational Commitment		
Objective Criteria:		
Salary		$\rho_{\text{pc}} = .16^{\text{dd}}$ ($k = 24$; $N = 10,012$) $\rho_{\text{pc}} = .06^{\text{ee}}$ ($k = 7$; $N = 2,495$) $\rho_{\text{pc}} = .16^{\text{vv}}$ ($k = 15$; $N = 8,267$)
Level (authority; promotion)	$\rho = .56^{\text{ff}}$ ($k = 6$; $N = 876$)	$\rho_{\text{pc}} = .04^{\text{dd}}$ ($k = 15$; $N = 6,984$) $\rho_{\text{pc}} = .13^{\text{vv}}$ ($k = 7$; $N = 3,060$)

See Table 9.1 footnotes.

TABLE 9.6B Meta-analytic Criterion-Related Validities of Compound Personality Variables (Self-Report) for Predicting Work Outcomes

Work Outcome (Criterion Construct)	Compound Personality Variable							
	Proactive Personality	Rugged Individualism	Creative Personality	Generalized Self-Efficacy	Violence	Sensation Seeking (Risk Taking)	Stress Tolerance	Big 5 R
Overall (Individual) Job Performance	$r = .21^{aa}$ ($k = 9$; $N = 1,272$)	$r = .05^p$ ($k = 32$; $N = 3,410$)		$\rho = .23^u$ ($k = 10$; $N = 1,122$)	$\rho_{rc} = -.41^{hh}$ ($k = 14$; $N = 4,003$)		$\rho_{rc} = .41^{hh}$ ($k = 13$; $N = 1,010$)	$\rho_c = .27^{kk/b}$
Overall (Team) Performance								$\rho_{rc} = .60^{kk/mm}$
Specific Occ. Success:								
Sales Effectiveness	$r = .29^{aa}$ ($k = 3$; $N = 959$)	$\rho_{rc} = .20^{uu}$ ($k = 5$; $N = 811$; ratings)						
Entrepreneurship—Business Creation			$\rho_{pc} = .24^{oo}$ ($k = 15$; $N = 4,620$)	$\rho_{pc} = .38^{oo}$ ($k = 8$; $N = 2,250$)		$\rho_{pc} = .10^{oo}$ ($k = 18$; $N = 8,863$)	$\rho_{pc} = .10^{oo}$ ($k = 6$; $N = 1,325$)	
Entrepreneurial Success	$\rho_{pc} = .27^{oo}$ ($k = 5$; $N = 679$)		$\rho_{pc} = .27^{oo}$ ($k = 7$; $N = 800$)	$\rho_{pc} = .25^{oo}$ ($k = 11$; $N = 1,331$)		$\rho_{pc} = .10^{oo}$ ($k = 13$; $N = 1,744$)	$\rho_{pc} = .20^{oo}$ ($k = 11$; $N = 1,282$)	
Expatriate Effectiveness								$\rho_c = .37^{kk/bb}$
Combat Effectiveness		$r = .25^p$ ($k = 2$; $N = 595$)						
Work Success Components:								
Contextual/Org. Citizenship (OCBs)—Overall								$\rho_c = .31^{kk/g}$
OCBs—General Compliance								$\rho_c = .23^{kk/l}$
OCBs—Altruism								$\rho_c = .13^{kk/l}$
Counterproductive Work Behavior (CWBs ¹)—Overall					$\rho_{rc} = .46^{hh}$ ($k = 4$; $N = 533$)		$\rho_{rc} = -.42^{hh}$ ($k = 5$; $N = 594$)	
CWBs ¹ —Interpersonal Deviance								$\rho_c = .44^{kk/d}$
CWBs ¹ —Org. Deviance								$\rho_c = .45^{kk/d}$
CWBs ¹ —Violence					$\rho_{rc} = .48^{hh}$ ($k = 11$; $N = 1,265$)			
Procrastination				$\rho = -.46^{ss}$ ($k = 39$; $N = 6,994$)		$\rho = .21^{ss}$ ($k = 11$; $N = 2,055$)		$\rho_{rc} = .70^{kk/ss}$
Workplace Safety (Accidents/Injuries—reversed)						$\rho_{pc} = -.20^h$ ($k = 3$; $N = 820$)		
Goal Setting								$\rho_{pc} = .63^x$ ($k = 4$; $N = 125$) $\rho_c = .54^{kk/x}$

Effort	$r = -.03^p$ ($k = 2$; N = 198)		
Getting Ahead			$\rho_{rc} = .27^{kk/o}$
Getting Along/Teamwork/ Interpersonal Effectiveness	$r = .06^p$ ($k = 4$; N = 306)		$\rho_{rc} = .33^{kk/o}$
Innovation & Creativity		$\rho_c = .37^a$ ($k = 15$; N = 1,086)	$\rho_{pc} = .32^{xx}$ ($k = 9$; N = ?)
		$\rho_{pc} = .17^{xx}$ ($k = 6$; N = ?)	
		$\rho_{pc} = .17^{yy}$ ($k = 5$; N = 1,031)	
Leadership & Leadership Effectiveness			$\rho_{pc} = .48^v$ ($k = 35$ -60; Ave. N = 200)
Leadership—Transformational			$\rho_c = .45^{kk/v}$
Knowledge, Education, Task, & Skill Criteria:			$\rho_c = .30^{kk/e}$
Task Performance/Proficiency	$r = .01^p$ ($k = 3$; N = 153)		
Educational Outcomes	$r = -.02^p$ ($k = 27$; N = 12,358)		
Training Outcomes (e.g., skill acquisition, grades)	$r = .03^p$ ($k = 11$; N = 1,614)		$\rho_c = .40^{kk/zz}$
Job & Career Satisfaction	$r = .17^{aa}$ ($k = 9$; N = 1,272)	$\rho = .45^u$ ($k = 12$; N = 12,903)	$\rho_{pc} = .41^w$ ($k = 38$ -92; Ave. N = 280)
	$\rho_{pc} = .38^{ee}$ ($k = 3$; N = 1,072)		$\rho_c = .39^{kk/ee}$
Objective Criteria:			
Tenure (Turnover—reversed)			$\rho = .35^{ww}$ ($k = 15$ -19; Ave. N = 96)
Salary	$\rho_{pc} = .11^{ee}$ ($k = 4$; N = 1,006)		

See Table 9.1 footnotes.

Whites, Blacks, Hispanics, Asians, and American Indians (Foldes et al., 2008; Hough et al., 2001), and one provided comparisons at the compound personality variables level (i.e., Integrity and Managerial Potential, a level even broader than the Big Five; Hough et al., 2001). In describing the differences we use the frequently used standard for definitions of small, medium, and large standardized mean differences (i.e., d s = 0.20, 0.50, and 0.80, respectively; Cohen 1988).

At the factor level (i.e., Emotional Stability, Extraversion, Conscientiousness, Agreeableness, and Openness to Experience), both meta-analyses concluded that Blacks, Hispanics, Asians, and American Indians score similarly to Whites, although sample sizes for the Asians and American Indians were often too small to have confidence in the stability of the findings. For those comparisons that had sample sizes of at least 1,500, absolute average d for both meta-analyses, combined, was 0.06, with differences ranging from approximately 0.00 to 0.20. Blacks and Hispanics scored higher than Whites about one-third of the time.

At the facet level, however, several differences emerge, with both meta-analyses providing similar results. We focus only on the Black, Hispanic, and White comparisons because sample sizes for the Asian and American Indian groups (ranging from 44 to 882) are too small to state conclusions with confidence. The data are interesting and complex.

Differences at the facet level of the FFM factor Extraversion are important. Sociability and Dominance are two facets of Extraversion. Although Blacks score about the same as Whites on Dominance, they score noticeably lower than Whites on Sociability ($d \approx 0.30$ to 0.40; Foldes et al., 2008; Hough et al., 2001). At the broad, factor level (i.e., Extraversion), Blacks score only slightly lower than Whites. At the global level of Extraversion, the facet-level differences are obscured; average group differences are reduced to about half the amount of the difference on the Sociability facet.

Some conclude that a result like this means that selection batteries should consist of measures of the global factor rather than measures of the facets. We argue the opposite but add a caveat. Our logic uses information about the criterion-related validity of the facets in conjunction with the knowledge that ethnic differences exist at the facet level.

We suggest that organizations take advantage of the meta-analytic evidence provided in Table 9.4 that indicates that Dominance correlates significantly with criteria such as overall job performance, sales effectiveness,

managerial effectiveness, getting ahead, innovation/creativity, educational outcomes, and level of authority, whereas Sociability correlates either negligibly, at a much lower level, or negatively with such criteria. Therefore, if the work analysis suggests that Dominance is a likely determinant of job performance, include a measure of Dominance (a facet-level measure) rather than a factor-level measure of Extraversion. Criterion-related validity is likely to be greater and adverse impact against Blacks is unlikely—a win-win outcome. Advice to practitioners to measure personality characteristics at the factor level may be ill-advised, although it is important to check gender and age comparisons to determine how such decisions will affect them.

Some differences emerge at the compound variable level (e.g., Integrity scales and Managerial Potential scales) and for measures of response bias (e.g., Unlikely Virtues scales). For Integrity scales, no differences between Whites and Blacks or between Whites and Hispanics reach even a small level of difference. For Managerial Potential scales, Blacks score on average about one-third of a standard deviation lower than Whites (Hough et al., 2001). No data were reported for other ethnic groups.

Of some importance is the difference between Whites and Hispanics on Unlikely Virtues scales. Hispanics score about one-half standard deviation higher than Whites on Unlikely Virtues (Hough et al., 2001). Although intentional distortion is discussed in a later section, we point out here that adjustments to content scale scores, such as scores on Conscientiousness and Emotional Stability, based on scores on Unlikely Virtues are potentially affecting Hispanics more than Whites, Blacks, and other ethnic groups. This potentially differential impact on Hispanics may inadvertently impact selection ratios of Hispanics.

Gender Group Similarities and Differences

A number of studies have examined gender differences on personality measures (e.g., Duehr et al., 2003; Else-Quest, Hyde, Goldsmith, & Hulle, 2006; Hough et al., 2001; Van Vugt, De Cremer, & Janssen, 2007). These studies come to similar conclusions: Women on average score higher than men on Dependability (a facet of Conscientiousness) and Agreeableness, whereas men on average score higher on Emotional Stability and Dominance (a facet of Extraversion). The differences range between approximately one-quarter of a standard deviation lower to approximately 0.4 standard deviation higher with women scoring about 0.4 standard deviation higher than men on Agreeableness and a quarter of a standard deviation higher on Dependability

(a facet of Conscientiousness) and a quarter of a standard deviation lower on Dominance (a facet of Extraversion) and Emotional Stability (Hough et al., 2001).

Age Group Similarities and Differences

A number of studies have examined age group differences in personality measures as well (e.g., Allemand, Zimprich, & Hendriks, 2008; Donnellan & Lucas, 2008; Hough et al., 2001; Roberts, Walton, & Viechtbauer, 2006; Terracciano, McCrae, Brant, & Costa, 2005). These studies come to similar conclusions: Differences between people age 40 years and older compared to people younger than 40 are minor, with the exception that older people score somewhat higher than younger people on Agreeableness and Conscientiousness and noticeably higher on the Dependability facet of Conscientiousness.

METHODS OF MEASURING PERSONALITY VARIABLES

In work settings, personality has typically been measured with self-report instruments using a Likert-type rating scale. The respondent is usually presented with a statement describing a behavior or attitude that reflects a high level of a particular trait, and the respondent indicates the extent to which that statement is true of him- or herself. These types of self-report measures are relatively easy to construct and have good levels of validity, as seen in Tables 9.1 through 9.6. The biggest problem with self-report measures is that they are susceptible to intentional distortion, or faking. Intentional distortion is discussed in detail in the next section. In this section, we focus on alternatives to traditional self-report personality measurement methods.

Forced-Choice Measures

Rather than asking respondents to describe themselves using a single stem with response options such as “strongly agree” to “strongly disagree,” forced-choice items present two statements and ask the respondent to choose the option that is most descriptive of him or her. Pairing statements that are about equal in their social desirability is intended to reduce faking (Edwards, 1953), although this seems to not necessarily be the case (e.g., Stanush, 1997). When the forced-choice scale presents the individual with statements representing two different traits, an undesirable

characteristic is ipsativity in the scores. Choosing one statement means not choosing the other statement, resulting in a higher score on one trait and a lower score on the other trait. Therefore, scale scores only provide information about the respondent’s trait level relative to his or her other traits, not about the respondent’s trait level relative to other people. This is a serious problem when using a personality measure for personnel selection or promotion, because decisions are made about people based on how they score relative to others. Ipsativity also forces a pattern of negative correlations between personality scales that would not otherwise be observed (Hicks, 1970) and distorts personality–criterion relationships. These effects can be diminished by using a large number of scales or by having respondents choose from a larger number of statements.

Computer Adaptive Personality Tests

Recently, forced-choice formats have been used to create computer adaptive personality tests. A computer adaptive test targets the trait level of the respondent with a smaller number of items than is necessary with a traditional personality measure. This allows for precise measurement, less testing time, and increased test security due to lower item exposure rates.

The measurement model on which these adaptive personality tests are based is an *ideal point* model, as opposed to the *dominance process* assumptions that are implicit in traditional personality measures (Stark, Chernyshenko, Drasgow, & Williams, 2006). The dominance process model is based on the assumption that people high on the trait being measured will tend to agree with positively worded items and people low on the trait will tend to disagree with those items. Therefore, traditional personality scales are constructed from items that represent extreme levels of traits.

Chernyshenko, Stark, Drasgow, and Roberts (2007) argued that, for personality measurement, the ideal point model is more appropriate than the dominance response process model. The ideal point model is based on Thurstone’s (1927) idea of using a paired comparison procedure to scale stimuli on more of an interval scale than is possible with a Likert-type response scale. Stark and Drasgow (1998) developed an algorithm to implement an adaptive measurement process using paired comparisons, and truly adaptive personality measurement was first applied with the development of the U.S. Navy’s Computer Adaptive Personality Scales (NCAPS; Houston, Borman, Farmer, & Bearden, 2005). In this type of test, the respondent selects which of two statements

representing different levels of the same personality trait is more descriptive of him or her. The next pair of statements presented is based on the trait-level estimate derived from previous responses. The selection of statement pairs is done in a way that maximizes the amount of item information at each step and continues until the trait estimate reaches an acceptable criterion of precision.

The NCAPS method of computer adaptive personality measurement does not suffer from the ipsativity problem associated with forced-choice inventories because the paired statements are always from the same trait. There is evidence that NCAPS-type measures are less susceptible to faking and have similar validities to traditional personality measures (Borman, Houston, McLellan, Schneider, & Kantrowitz, 2011). This process has now been used to develop a commercially available computer adaptive personality test that is applicable across a wide variety of jobs, levels, and industries (PreVisor, 2010), and this type of test has now been developed for and implemented by the Office of Personnel Management and the Department of Defense for selection purposes.

Another type of computer adaptive personality test that is based on a different measurement model is the Tailored Adaptive Personality Assessment System (TAPAS; Stark, Drasgow, & Chernyshenko, 2008). TAPAS uses a multidimensional pairwise preference (MDPP) format that is designed to promote resistance to faking. When forming item pairs, TAPAS balances two statements representing different traits in terms of social desirability and extremity on the traits they assess. The ipsativity issue was solved by adding a small number of unidimensional item pairs in with the multidimensional item pairs (i.e., the MDPP items). The unidimensional pairs are needed to identify the latent trait metric and yield normative scores using the MDPP format (Stark, 2002; Stark, Chernyshenko, & Drasgow, 2005). TAPAS scoring is based on the MDPP IRT model originally proposed by Stark (2002). A series of equations are solved numerically to produce a vector of latent trait scores for each respondent as well as standard errors. TAPAS measures a trait taxonomy comprising 22 facets of the Big Five personality factors plus Physical Conditioning. A 13-facet version of TAPAS has been implemented in military entrance processing stations.

Conditional Reasoning Tests

Conditional reasoning tests have the appearance of a traditional reasoning test, but they are actually intended to measure personality characteristics. Conditional reasoning tests have been created to measure aggression

(James, 1998) and achievement (James, McIntyre, Glisson, Bowler, & Mitchell, 2004). This approach is based on the idea that individuals with undesirable traits use ego-protective cognitive biases to justify their actions. For example, aggressive individuals are more likely to perceive hostile intent in others (referred to as hostile attribution bias), making their aggressive behavior seem justifiable to them. Items present scenarios that prime these justification mechanisms, and the response options present logically correct responses that differ in terms of personality-relevant motives. The respondent receives a point if the response option keyed to the target personality trait is chosen. According to a meta-analysis, the Conditional Reasoning Test of Aggression produces useful uncorrected validities (e.g., $\bar{r} = .26$ for predicting counterproductive work behavior; Berry, Sackett, & Tobares, 2010). Conditional reasoning measures tend to have small correlations with self-report measures and are easily faked when respondents are aware that the items are measuring personality variables.

Other Ratings

A simple variation of a traditional personality measure is to solicit personality ratings from others (e.g., coworkers) who can describe a target individual's personality. Use of other ratings is rare in I-O psychology, but it is common in basic personality research, where well-acquainted observers have been found to be quite accurate in rating targets (Connelly & Ones, 2010; Funder, 1995). Nevertheless, there is strong empirical support showing that personality variables measured with observers' ratings have good validity for predicting job performance (Mount, Barrick, & Strauss, 1994; Tupes, 1957, 1959). A meta-analysis showed that other ratings produced significantly higher criterion-related validities than did self-ratings (Connelly & Ones, 2010). This suggests that observers may have a more accurate view of targets than targets have of themselves. It is also possible to collect personality ratings from multiple observers, which allows for the potential of very high validity coefficients when multiple ratings are combined into a more reliable average.

Of course, there are pragmatic issues associated with collecting observers' ratings in personnel selection settings. No studies have examined observer ratings in a selection context, so it is unclear whether observers' reports are more or less susceptible to faking and adverse impact (particularly from stereotypes and prejudices potentially held by raters). In addition, it would be difficult to find a reasonable sample of raters for individuals

who are not already part of the organization. Further research addressing these pragmatic issues for selection is needed.

INTENTIONAL DISTORTION

One issue that still haunts the use of personality variables in high-stakes testing in work (and educational) settings is intentional distortion, or faking. Research since the first edition of this handbook has produced important findings and new knowledge that has significant implications for practice.

There are important findings that have stood the test of time. We knew them when the first edition was published, and they have not changed, although research has produced greater understanding. Following is a summary of what we knew then, what we know now, and about what we want to know more (see Hough & Connelly, in press; Hough & Oswald, 2008, for more detail):

- Test takers can, when instructed to do so, distort their responses in a positive direction and even more dramatically in a negative direction, and social desirability scales are sensitive to such intentional distortion (Hough et al., 1990; Viswesvaran & Ones, 1999).
- Intentional distortion is larger in directed faking studies than in real-life, high-stakes testing situations (Hough, 1998; Viswesvaran & Ones, 1999).
- Studies that examine factor structure in real-life applicant settings indicate that the factor structures of applicant responses do not show the same collapsed structure that is observed in instructed faking studies (e.g., Ellingson, Smith, & Sackett, 2001; Montag & Comrey, 1990; Schmitt & Ryan, 1993).
- In high-stakes, real-life testing situations, relationships between content personality scales (such as Conscientiousness) and performance outcomes do not appear to be appreciably affected by social desirability scale scores, at least when measured with existing social desirability scales (Hough, 1998; Hough & Ones, 2001; Hough et al., 1990; Ones, Viswesvaran, & Reiss, 1996; Schmitt & Oswald, 2006).
- Considerable evidence indicates that, even in high-stakes testing situations, intentional distortion appears not to be corrosive of criterion-related validity; predictive validity studies with applicants indicate that criterion-related validities seem unaffected. For example, Ones et al. (1993) meta-analyzed criterion-related validities of personality-based integrity tests (which are saturated with Conscientiousness, Agreeableness, and Emotional Stability), finding that they correlated with broadly defined counterproductive behavior .29 (corrected; sample size equaled 93,092; number of studies equaled 62).
- Nonetheless, faking issues are of paramount concern in high-stakes testing situations where test takers are motivated to obtain scarce resources such as jobs, scholarships, or school admissions. These typically are top-down selection situations in which people who have distorted their self-descriptions rise to the top of the distribution and are among the first to be selected. The situation is more problematic in low selection ratio settings where only a small percentage of the applicant pool is selected (Mueller-Hanson, Heggstad, & Thornton, 2003; Zickar, Rossé, Levin, & Hulin, 1996).
- Applicants do tend to score higher than incumbents but it is not as large as often thought. A meta-analysis comparing incumbents with non-incumbents found moderate standardized mean score differences on Emotional Stability and Conscientiousness but very small differences on other Big Five factors (Birkeland, Manson, Kisamore, Brannick, & Smith, 2006).
- There are, however, wide variations and differing conclusions across studies that have examined mean score differences between applicants and incumbents. Comparisons need to take moderating variables, such as personality characteristic, job type, and type of test (Birkeland et al., 2006), into account.
- Within-person studies have also found widely differing results in applicant compared to incumbent settings. Examples of studies that have found large differences are Griffith, Chmielowski, and Yoshita (2007) and Lönnqvist, Paunonen, Tuulio-Henriksson, Lönnqvist, and Verkasalo (2007). An example of a study that did not find large differences is Ellingson, Sackett, and Connelly (2007). A potential explanation for these differences is the existence of warnings not to fake.
- Warnings not to fake and consequences for faking appear to diminish the expected effects of high-stake testing situations. The Birkeland et al. (2006) meta-analysis did not and probably could not examine the moderating effects of warnings and consequences. However, results from one large-scale study involving three organizations and many thousands of applicants suggest that intentional distortion is minimized when test takers are warned not to distort their self-descriptions (e.g., Hough, 1998).

- Meta-analytic evidence indicates that warnings along with consequences for faking reduce distortion; warnings without consequences do not (Dwight & Donovan, 2003). Another meta-analysis indicates that warnings are effective deterrents against distortion in real-life applicants although they are ineffective in faking studies (Stanush, 1997).
- Although mean score differences between those who are warned versus those who are not warned exist, warnings appear not to affect criterion-related validity of personality variables; validities remain intact (Converse et al., 2008; Fox & Dinur, 1988; Robson, Jones, & Abraham, 2008).
- The framing of a warning message, positive versus negative, may have differential effects on mean scores (Converse et al., 2008).
- Clearly, norms against which to compare applicants for whatever is at stake (e.g., job, school admission) should be based on applicant data that was gathered using the same set of test instructions.
- Warnings appear to affect scores on Conscientiousness and their relationship to cognitive ability (Vasilopoulos, Cucina, & McElreath, 2005). The extent to which Conscientiousness correlates with cognitive ability reduces its usefulness as an alternative predictor of performance outcomes.
- Coaching can improve test taker ability to avoid detection of faking on traditional social desirability scales (Hurtz & Alliger, 2002).
- Subtle items appear more resistant to faking than are more obvious items (Alliger, Lilienfeld & Mitchell, 1996; White, Young, Hunter, & Rumsey, 2008).
- Corrections to content scales, such as Conscientiousness, based on traditional social desirability scale scores, do not affect relationships between content scales and criteria (Hough, 1998; Schmitt & Oswald, 2006).
- One of the criticisms of the faking literature is the use of traditional social desirability scales to detect and correct for faking. Traditionally, personality test developers have written items for social desirability scales that consist of unlikely virtues—behaviors, attitudes, and characteristics that, if endorsed, suggest the person is describing him- or herself in an overly virtuous way, rendering responses to other items suspect as well.
- Researchers have developed new types of scales to detect socially desirable responding. One promising strategy uses idiosyncratic patterns of item responses (Kuncel & Borneman, 2007). Faking scales are based

on subsets of items that show consistent multimodal distributions when respondents are instructed to fake. The large differences between honest and faked distributions are used to identify items that more accurately differentiate honest respondents from those who are faking.

In the past decade, models of faking have emerged that incorporate more sophisticated thinking. One model of faking includes attitudes toward faking, perceived control, and group norms for faking as variables that determine faking. Depending on moderating conditions of warnings and ability to fake, these variables predict faking (McFarland & Ryan, 2000, 2006). Another more recent model incorporates many of these individual difference variables as well as contextual, moderating factors (Goffin & Boyd, 2009). These models, along with newer process models of the personality determinants of work behavior, offer considerable promise.

We suggest continued research in developing models of faking. We also suggest that future research focus on preventing or reducing faking. We need better understanding of the effects of (a) warnings not to fake, types of warnings, and how and when warnings are communicated; (b) consequences for faking and types of consequences; (c) asking follow-up questions; (d) the form of follow-up questions, such as written elaboration, interview with respondent, interview with friends, family, and/or coworkers; (e) reactions of test takers to warnings and consequences; (f) subjective versus factual items on faking; and (g) subtle versus obvious items (items that are transparent or easy to discern what personality characteristic is being measured). We also need to understand coaching and its effects.

We are, however, leery of recommending more experimental, laboratory studies. There are too many instances of such results not generalizing to real-life settings. Many of the findings based on laboratory experiments with college students have mired us in unfruitful discussions. Of course, there will be questions for which such populations in such settings can provide useful information, but researchers need to ask very pointed questions about the likely utility of another lab study with college students.

SUMMARY AND PATH FORWARD

In the past decade, we have added considerably to our storehouse of knowledge on the relationships between specific personality facets and a variety of different

criteria. The usefulness of personality variables for predicting organizationally relevant criteria is now well established. It is now time to cumulate our knowledge in a more systematic way by making a concerted effort to collect data and organize results according to a standard taxonomy of criteria that more adequately represents the individual job performance domain. When predictors and criteria are more standard, this will allow meta-analyses to be conducted and a synthetic validation database to be developed as quickly as possible. This database will allow us to use synthetic validation techniques to estimate the validity of a battery of personality variables for any job that includes performance dimensions on which research is available (Johnson et al., 2010). The database will also advance science by greatly increasing our knowledge base with respect to relationships between different personality and criterion variables. This will not only enhance the science of personnel selection, but will be a rich source of data for personality process research (e.g., to create a meta-analytic correlation matrix as input to a structural equation model; Viswesvaran & Ones, 1995).

The past decade has also seen great advances in personality measurement methods. Computer adaptive personality tests allow for greater precision in measurement in a shorter amount of testing time, while evidence suggests that validity is enhanced and intentional distortion is managed to some extent. Because test security is enhanced due to each applicant only seeing a small subset of the items, this type of measure is ideal for unproctored Internet testing, which is becoming increasingly common. Conditional reasoning tests have promise for reducing intentional distortion while maintaining high validity for certain constructs, as long as test takers are unaware that the test measures personality. Another measurement method that deserves additional research is personality ratings made by knowledgeable observers. If multiple such observers are available, the enhanced reliability has the potential to yield validities at levels not seen with typical self-report measures.

Finally, the past decade has seen greatly increased interest in explaining the mechanisms through which personality influences job performance by developing process models. Johnson and Hezlett (2008) reviewed a number of process models and developed a general model demonstrating the many variables through which personality can influence performance. Personality can influence performance through job knowledge acquisition, skill acquisition, different components of motivation, and attitudes, among other variables. Personality will influence

performance through different variables depending on the type of performance and the situation. Further research on process models will be necessary to gain a greater understanding of the relationships between different personality traits and criteria of interest to organizational researchers. We are confident that the next decade will see an even greater leap forward in our understanding of personality in work settings than we saw in the past decade.

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